

Fantasy Cricket Platform

Complete Technical Architecture & Functional Specification Report

PRODUCTION GRADE
V2.4.0 · 2026 RELEASE

Full-Stack SSR

TANSTACK START + NODE.JS

Real-Time Feed

SOCKET.IO LIVE WEBSOCKETS

Dual 2FA OTP

NODEMAILER SMTP + MONGO
ATLAS

Isolated Ledger

100% PER-USER WALLET
PASSBOOK

1. Executive Overview & Technology Stack

PRODUCTION READY

An enterprise-grade, real-time sports gaming web platform simulating the complete lifecycle of competitive fantasy cricket. Features full server-side rendering, ball-by-ball simulated cricket engine, instant user wallet operations with double-entry passbooks, and dual admin governance controls.

Frontend Architecture

TanStack Start SSR · React 18 · Vite ·
Tailwind CSS · Lucide Icons · Radix UI ·
Socket.IO Client.

Backend Services

Node.js · Express · TypeScript · MongoDB
Atlas (Mongoose) · JWT Authentication ·
Nodemailer SMTP.

Cloud & DevOps

Docker Containerization · Nginx Reverse
Proxy · AWS EC2 Ubuntu Production Node
· Live Socket WebSockets.

2. Authentication, 2FA OTP & Identity Protection

SECURITY LAYER

Registration & Dual-Collection 2FA

- Registration:** Collects user name, email, phone number, and salt-hashed password (bcrypt).
- Plain-Text 6-Digit OTP:** Dispatched via Nodemailer SMTP to user's registered inbox upon sign-up and login.
- Dual-Collection Persistence:** Mirrored to both otps and otp collections in MongoDB Atlas, with dropped zero-second TTL indexes to prevent unexpected auto-deletion.
- 10-Minute Expiry:** Validates strictly within 600s with anti-replay state checks.

Session Security & Access Controls

- JWT Authorization:** Stateless bearer tokens signed with secret key for API route protection.
- Resend OTP Safeguards:** Cooldown timer (60s) with rate-limiting to prevent spam abuse.
- Role-Based Access (RBAC):** Strict separation between standard players, Tournament Admins, and Super Administrators.
- Password Recovery:** OTP verification triggers secure tokenized password reset flow.

3. Live Match Center & Cricbuzz-Style Cricket Desk

REAL-TIME CORE

Cricbuzz-Inspired Experience

- Global Ticker Strip:** Live horizontal scrolling bar displaying all active world matches with team flags, live scores, and pulsing status.
- Rotating Cricket News Desk:** Time-shifted breaking world cricket headlines on a dedicated 3-column dashboard.
- Featured Video Showcase:** Curated match highlights and tactical analysis previews with embedded duration badges.

Detailed Match Scorecard & Ball-by-Ball

- Split-Table Layout:** Real-time batsman table (R, B, 4s, 6s, SR) and bowling metrics (O, M, R, W, Economy).
- Key Stats Sidebar:** Current partnership runs/balls, last fallen wicket, remaining overs, and toss results.
- Over Summary Strip:** Recent ball events breakdown (e.g., 0 · 4 · 1 · W · 6 · 0) updated live.

4. Fantasy Team Creation & Rules Validation Engine

GAME LOGIC

Strict mathematical validation guarantees competitive balance and follows standard international fantasy sports guidelines:

Rule Category	Constraint	Validation Logic & Edge-Case Protection
Squad Size	Exactly 11 Players	Save action disabled until exactly 11 valid players are selected.
Salary Cap	Max 100 Virtual Credits	Each player costed (7.5 - 11.0 cr). Exceeding 100 credits blocks submission.
Team Balance	Max 7 from One Real Team	Prevents one-sided team stacking; enforces multi-team selection.
Role Quotas	WK: 1-4 BAT: 3-6 AR: 1-4 BOWL: 3-6	Dynamic role counters alert user to minimum and maximum composition limits.
Multiplier Boosters	Captain (2.0x) & Vice-Captain (1.5x)	Mandatory distinct captain and vice-captain selection before saving team.

5. Contests Arena, Leagues & Private Competitions

COMPETITION ENGINE

Contest Formats

- Mega Contests:** Multi-entry pools with tiered rank-based prize distributions.
- Head-to-Head (1v1):** High-stakes head-to-head showdowns with winner-take-all mechanics.
- Winner Takes All:** 3-10 player mini-leagues where the top scorer claims the total pool.
- Practice Contests:** Free zero-risk entry for onboarding new players without monetary stake.

Private Contest Creation

- Custom League Creation:** Users create private tournaments by defining name, total spots, and entry fees.
- Unique Invite Codes:** Automatic alpha-numeric code generation for private circle sharing.
- Auto Fee Deductions:** Instant entry fee deduction from user's isolated wallet upon joining.
- Lock on Match Start:** Automatic contest freezing when match live simulation commences.

6. Persistent User-Specific Wallet & Audit Passbook

FINANCIAL ISOLATION

100% User Data Isolation

- Per-User Balances:** User A, User B, and User C maintain separate database records in MongoDB.
- New User Defaults:** Newly registered accounts initialize strictly with ₹0.00 balance.
- Header & Page Parity:** Top navigation bar badge matches the dedicated /wallet screen in real time.
- Session Persistence:** Balances persist accurately across logouts, logins, and browser refreshes.

Passbook & Withdrawal Safeguards

- Add Cash (Instant Credit):** Simulated UPI / Card payment gateway with instant wallet credit.
- ₹100 Maintenance Reserve:** Prevents total wallet depletion by enforcing a ₹100 minimum operating balance.
- Double-Entry Passbook:** Every transaction logs Type (DEPOSIT, WITHDRAWAL, CONTEST_ENTRY, WINNING), Amount, Status, and Timestamp.
- Atomic Operations:** Prevents balance corruption during concurrent entry fee deductions.

7. Real-Time Scoring Engine & Live Leaderboards

DYNAMIC CALCULATION

Comprehensive Point System

- Batting:** Run (+1 pt), Boundary 4 (+1 bonus), Six (+2 bonus), 30 Runs (+4 bonus), 50 Runs (+8 bonus), Century (+16 bonus), Duck dismissal (-2 pts).
- Bowling:** Wicket (+25 pts, excl. run-out), Bowled/LBW (+8 bonus), 3-Wicket haul (+4 bonus), 4-Wicket (+8 bonus), 5-Wicket (+16 bonus), Maiden over (+12 pts).
- Fielding:** Catch (+8 pts), Stumping (+12 pts), Direct Run-Out (+12 pts).
- Multipliers:** Captain receives 2.0x points, Vice-Captain receives 1.5x points.

Live Leaderboard Rank Shifting

- Real-Time Broadcast:** Socket.IO emits recalculations as each ball is bowled in the match simulator.
- Instant Rank Shifting:** Leaderboard dynamically sorts participant fantasy teams by total points.
- Breakdown Drawer:** Click on any participant to view their 11-player squad and individual point contributions.
- Prize Settlement:** Automated prize distribution into winners' wallets once match reaches COMPLETED status.

8. Admin & Super Admin Governance Console

SYSTEM CONTROL

Match Lifecycle Control

Schedule matches, toggle status (Upcoming → Live → Completed), manually trigger or pause ball-by-ball simulation, and manage team rosters.

Contest Management

Publish public contests, set pool sizes and entry fees, cancel void matches with auto-refunds, and oversee prize settlement payouts.

User & Wallet Auditing

Inspect all registered users, review full transaction ledgers, monitor suspicious activities, and adjust wallet balances with audit logs.

9. Production Infrastructure & Verification Status

DEPLOYED & VERIFIED

Subsystem	Implementation Detail	Status
User Authentication & 2FA	MongoDB otps + otp dual-write with 10-minute expiry and Nodemailer SMTP delivery.	100% VERIFIED
Per-User Wallet Engine	Strict user ID isolation, ₹100 maintenance reserve, instant passbook tracking.	100% VERIFIED
Cricbuzz Cricket Portal	Live match ticker, rotating international news desk, batsman & bowler tables.	100% VERIFIED
Fantasy Team Validation	100 credit ceiling, 11 player requirement, role quota guards, C/VC multipliers.	100% VERIFIED
Cloud Deployment	AWS EC2 Linux host running Docker containers, Nginx reverse proxy, PM2 process manager.	LIVE ON EC2